Supplementary Material

[CatBack: Universal Backdoor Attacks on Tabular Data via Categorical Encoding]

About this file. This PDF contains the full text of the section that was moved to the supplementary material for the CatBack paper.

1 Spectral Signatures Results

In this section, we have included the results for the Spectral Signature for target label 1 on all datasets and models. We observed that the defense was most successful in this setting compared to other settings. As mentioned in the paper, the defense is mostly successful in detecting the poisoned samples from the FTT model on the BM dataset. There are also some distinctions between CovType on FTT and HIGGS on Saint, but even for them, decreasing the μ value will rapidly result in the defense, decreasing the scores for poisoned samples and failure. When μ is smaller, fewer samples are used for the trigger calculation. Most of these samples are close to the decision boundary, resulting in a trigger of smaller magnitude, making the attack stealthier. This could also affect the ASR.

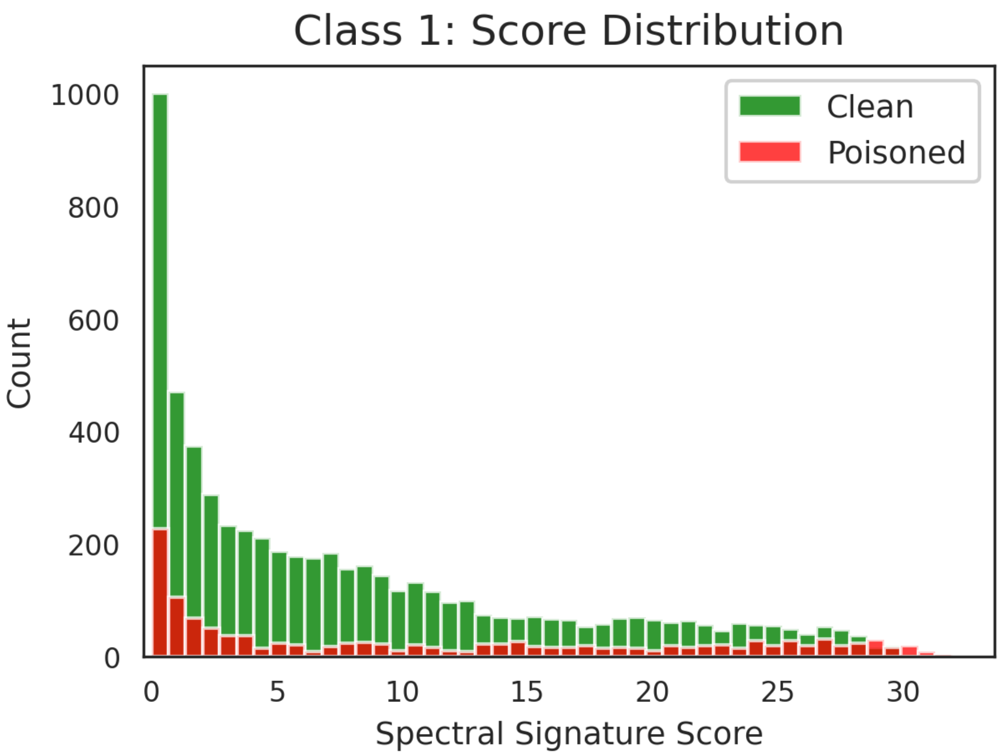

Figure 1: Spectral Signatures: target class=1, Model: FTT, Dataset: ACI, $\epsilon = 0.05$, and $\mu = 1.0$).

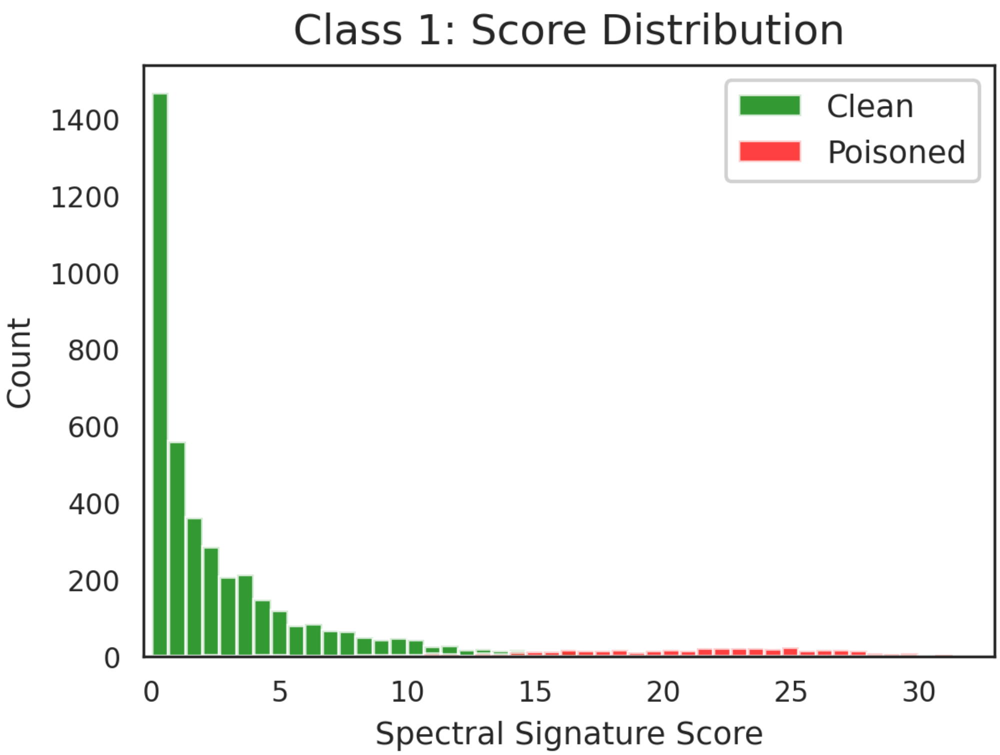

Figure 2: Spectral Signatures: target class=1, Model: FTT, Dataset: BM, $\epsilon = 0.05$, and $\mu = 1.0$).

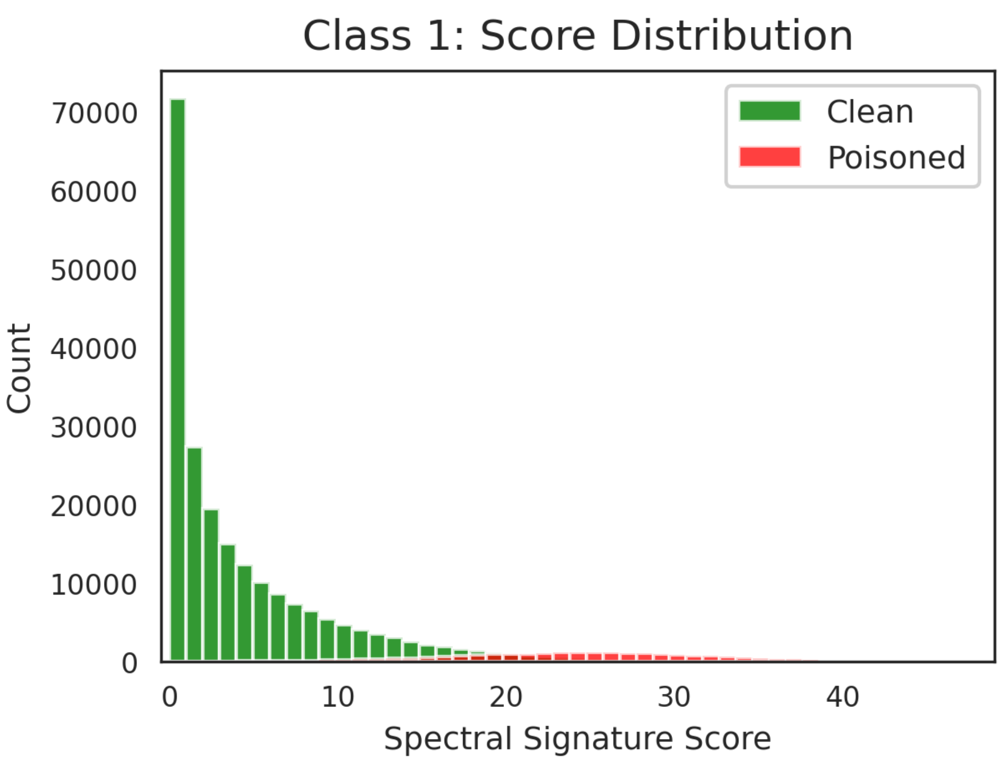

Figure 3: Spectral Signatures: target class=1, Model: FTT, Dataset: CovType, $\epsilon = 0.05$, and $\mu = 1.0$).

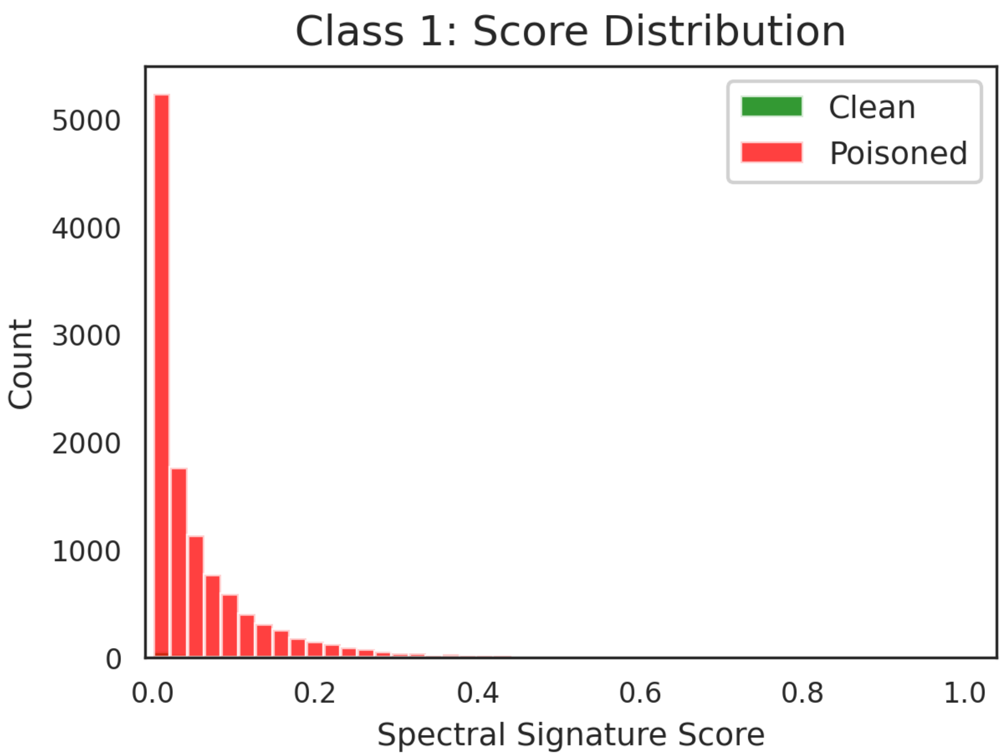

Figure 4: Spectral Signatures: target class=1, Model: FTT, Dataset: Credit Card, $\epsilon = 0.05$, and $\mu = 1.0$).

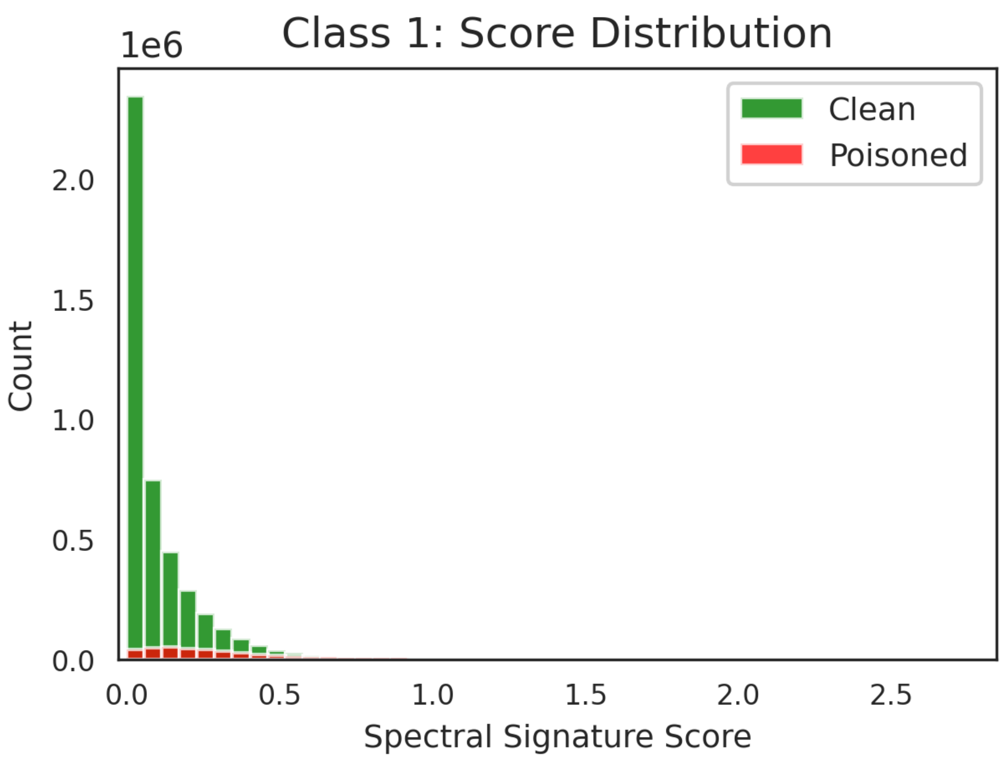

Figure 5: Spectral Signatures: target class=1, Model: FTT, Dataset: HIGGS, $\epsilon = 0.05$, and $\mu = 1.0$).

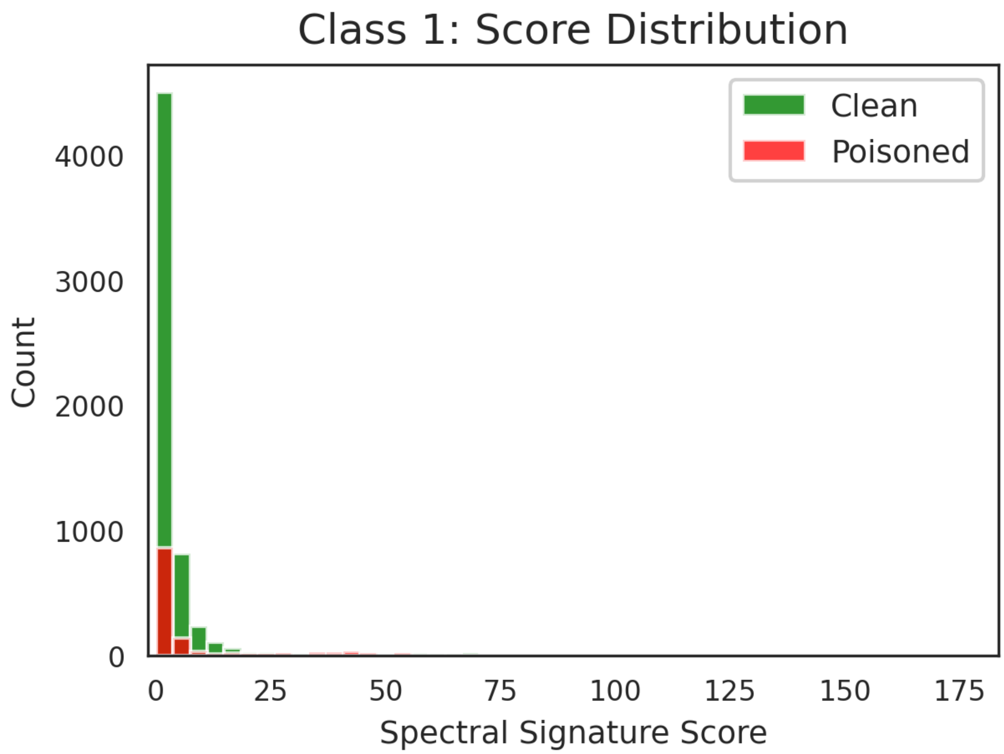

Figure 6: Spectral Signatures: target class=1, Model: TabNet, Dataset: ACI, $\epsilon = 0.05$, and $\mu = 1.0$).

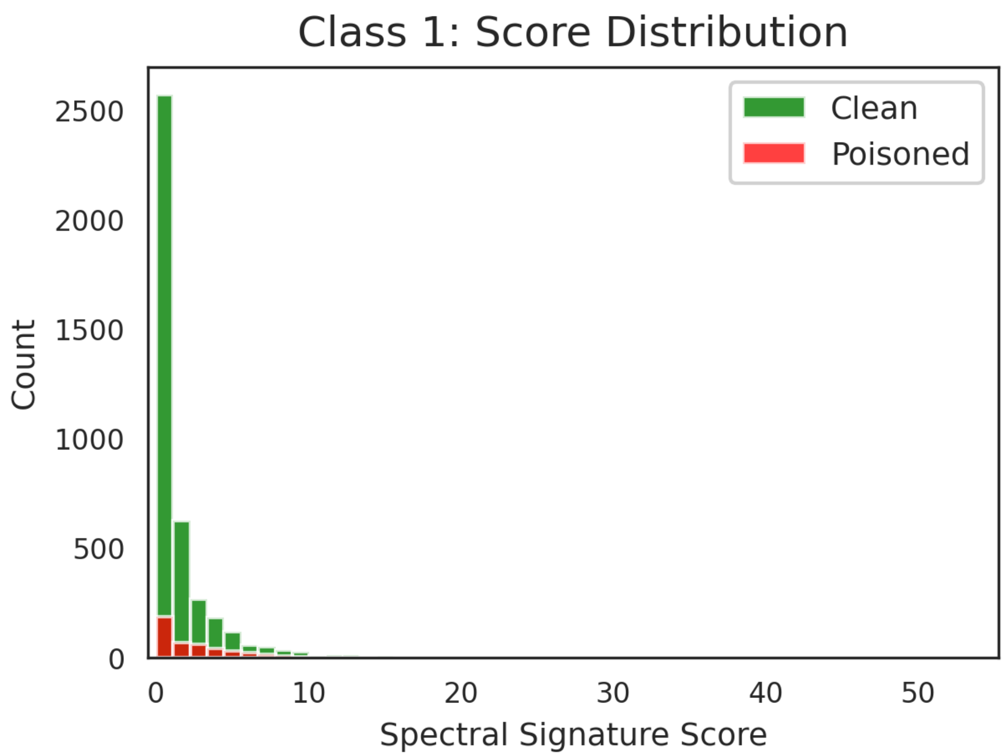

Figure 7: Spectral Signatures: target class=1, Model: TabNet, Dataset: BM, $\epsilon = 0.05$, and $\mu = 1.0$).

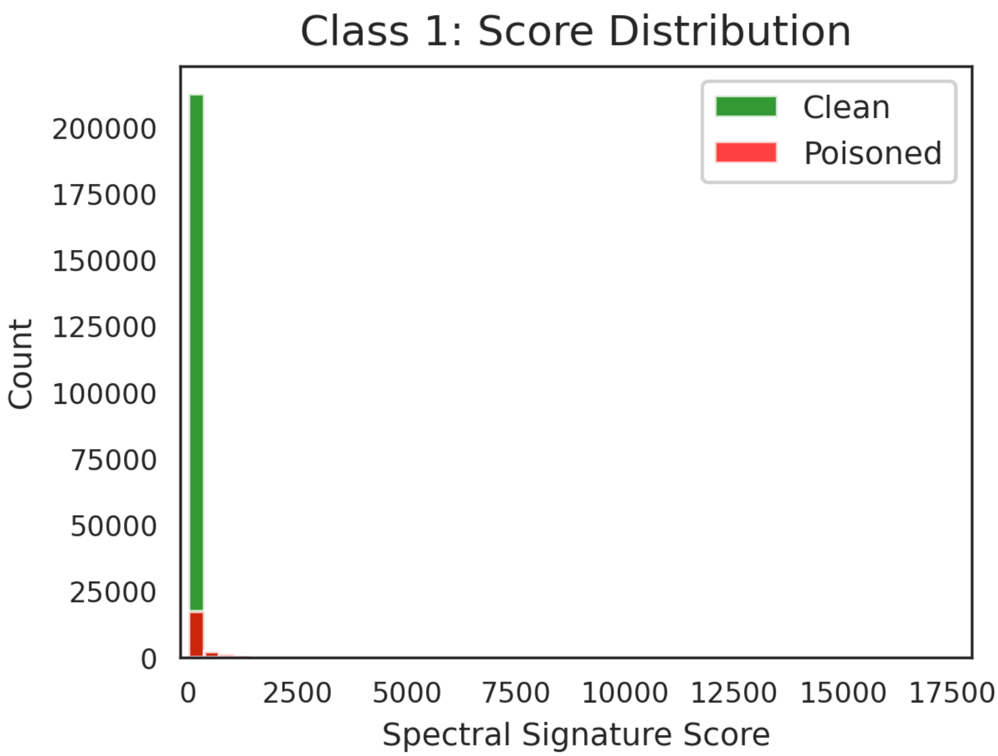

Figure 8: Spectral Signatures: target class=1, Model: TabNet, Dataset: CovType, $\epsilon = 0.05$, and $\mu = 1.0$).

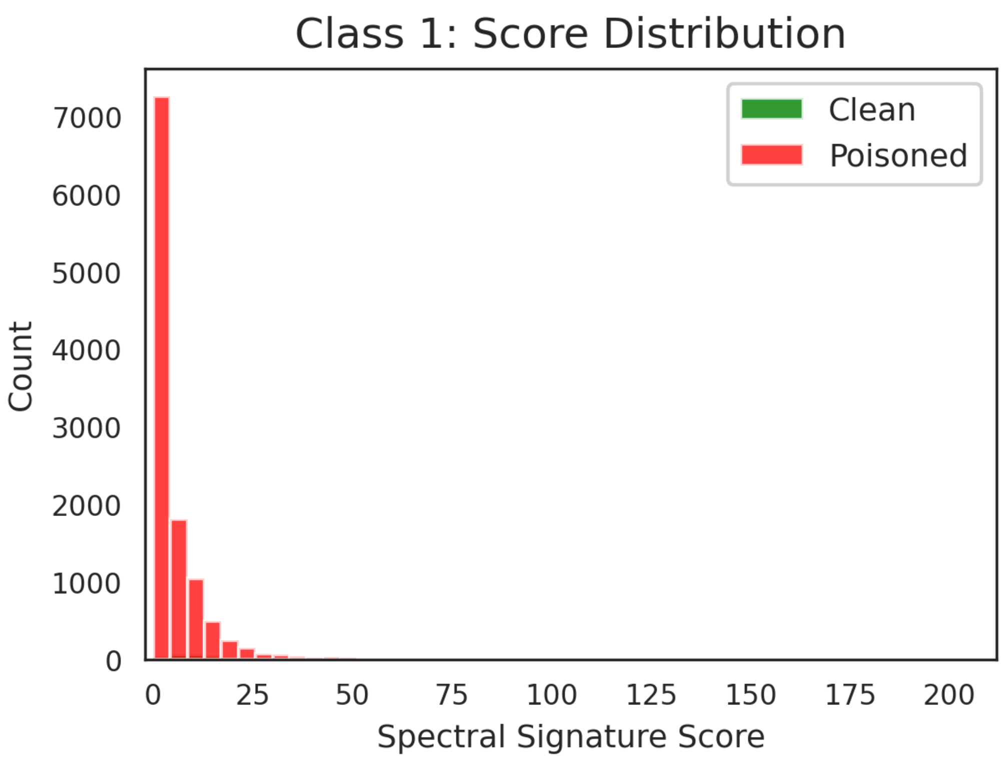

Figure 9: Spectral Signatures for: target class=1, Model: TabNet, Dataset: Credit Card, $\epsilon = 0.05$, and $\mu = 1.0$).

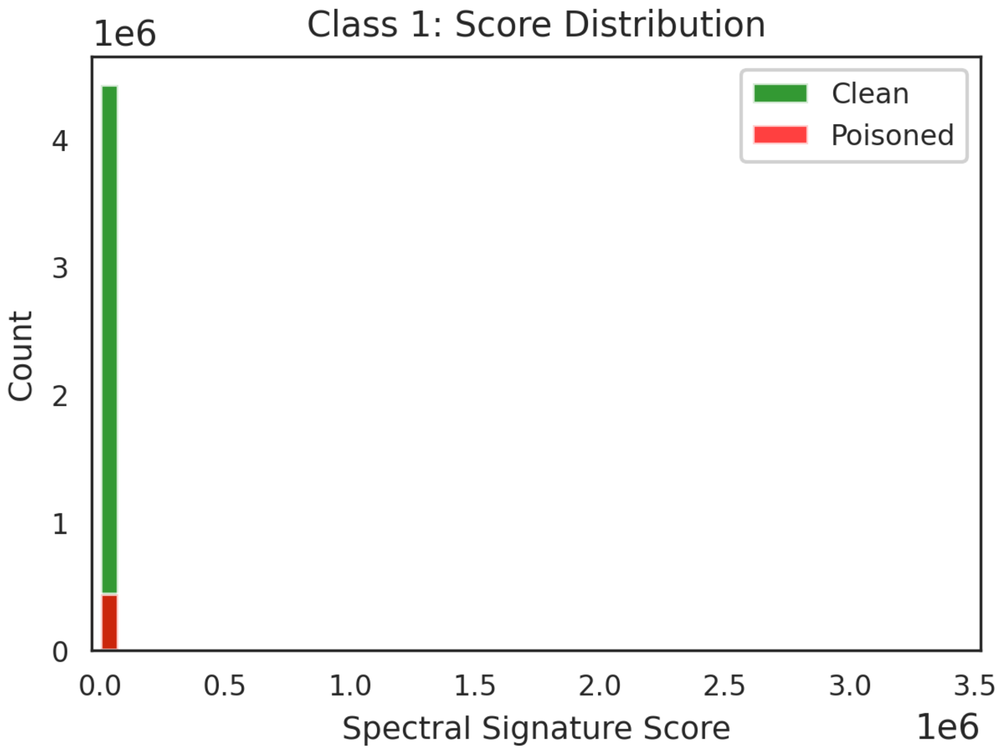

Figure 10: Spectral Signatures: target class=1, Model: TabNet, Dataset: HIGGS, $\epsilon = 0.05$, and $\mu = 1.0$).

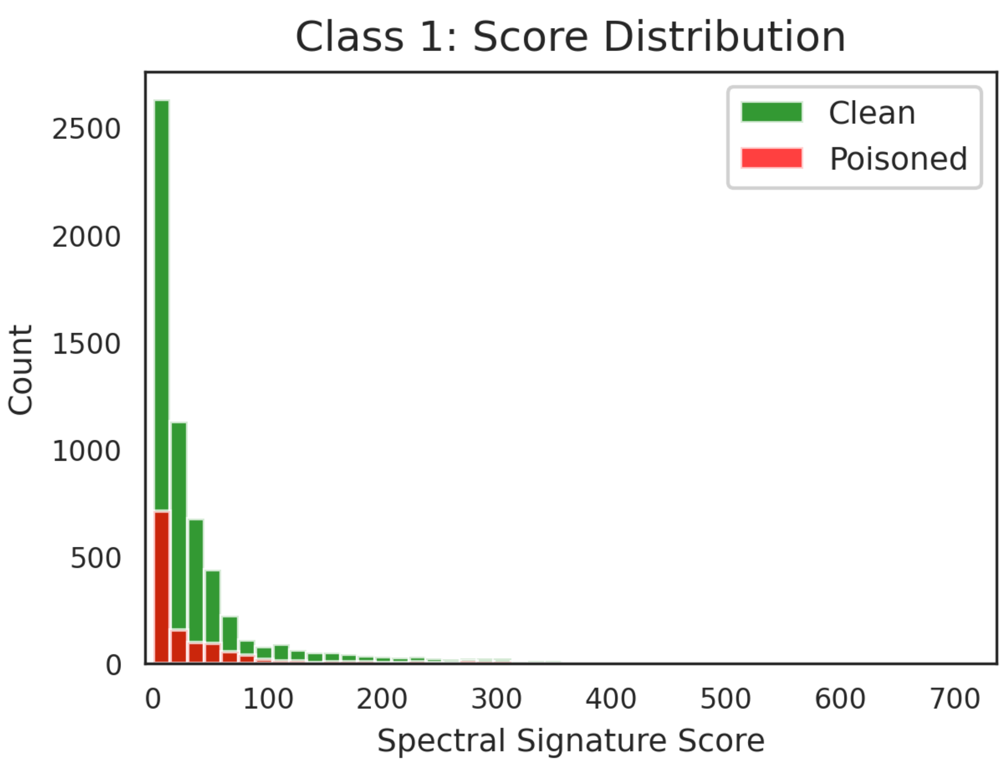

Figure 11: Spectral Signatures: target class=1, Model; Saint, Dataset: ACI, $\epsilon = 0.05$, and $\mu = 1.0$).

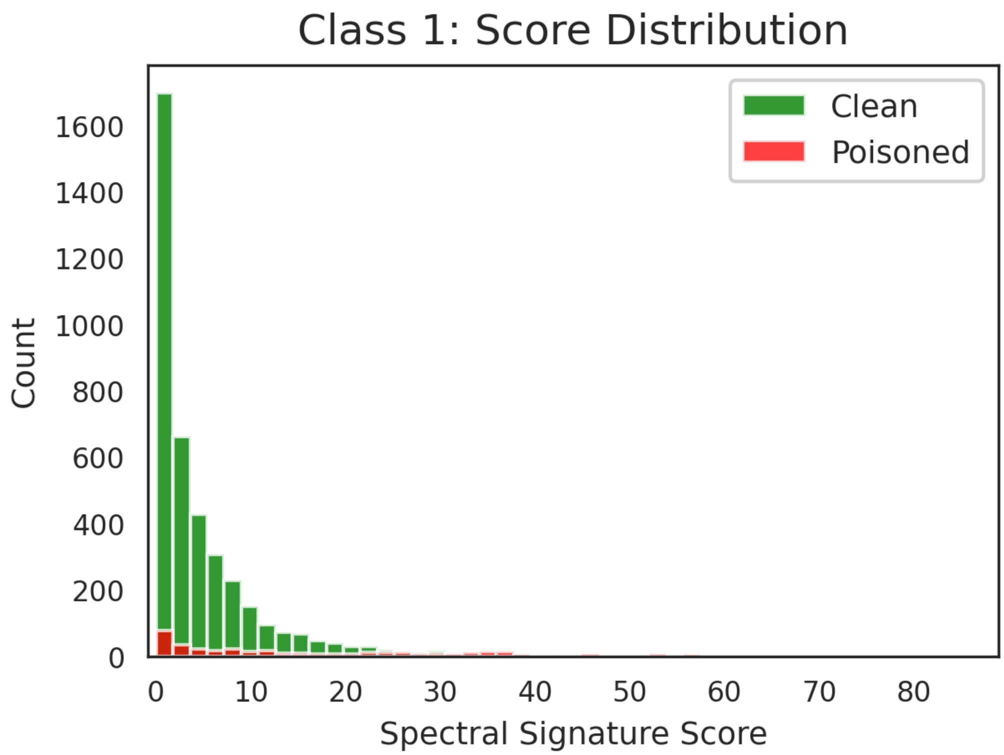

Figure 12: Spectral Signatures: target class=1, Model: Saint, Dataset: BM, $\epsilon = 0.05$, and $\mu = 1.0$).

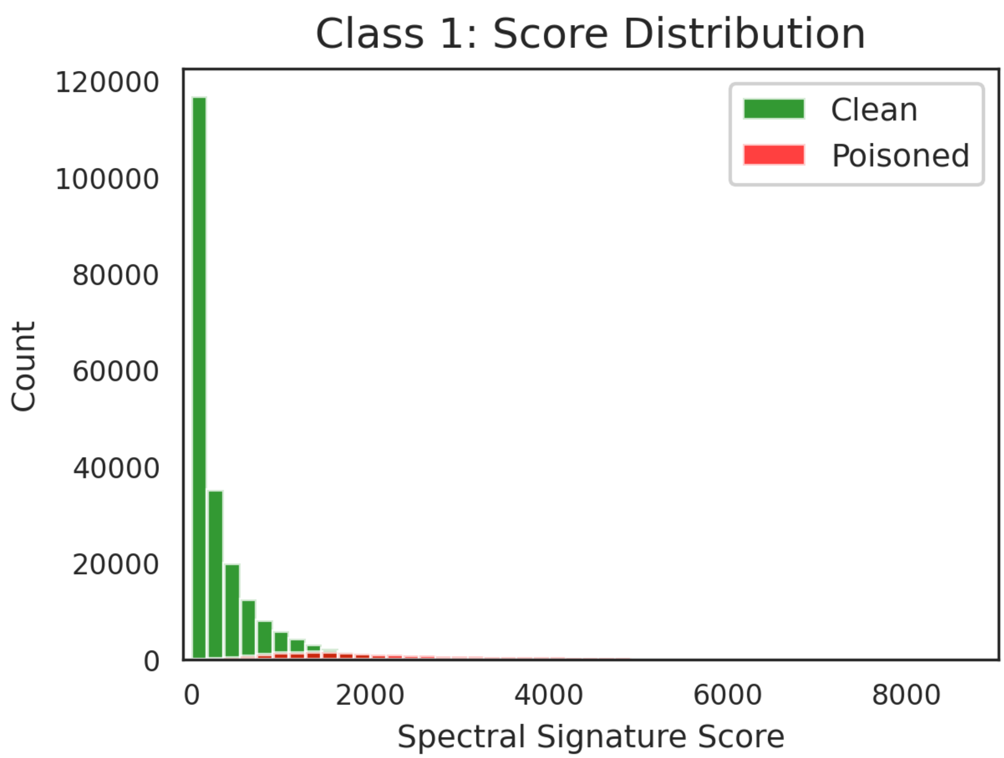

Figure 13: Spectral Signatures: target class=1, Model: Saint, Dataset: CovType, $\epsilon = 0.05$, and $\mu = 1.0$).

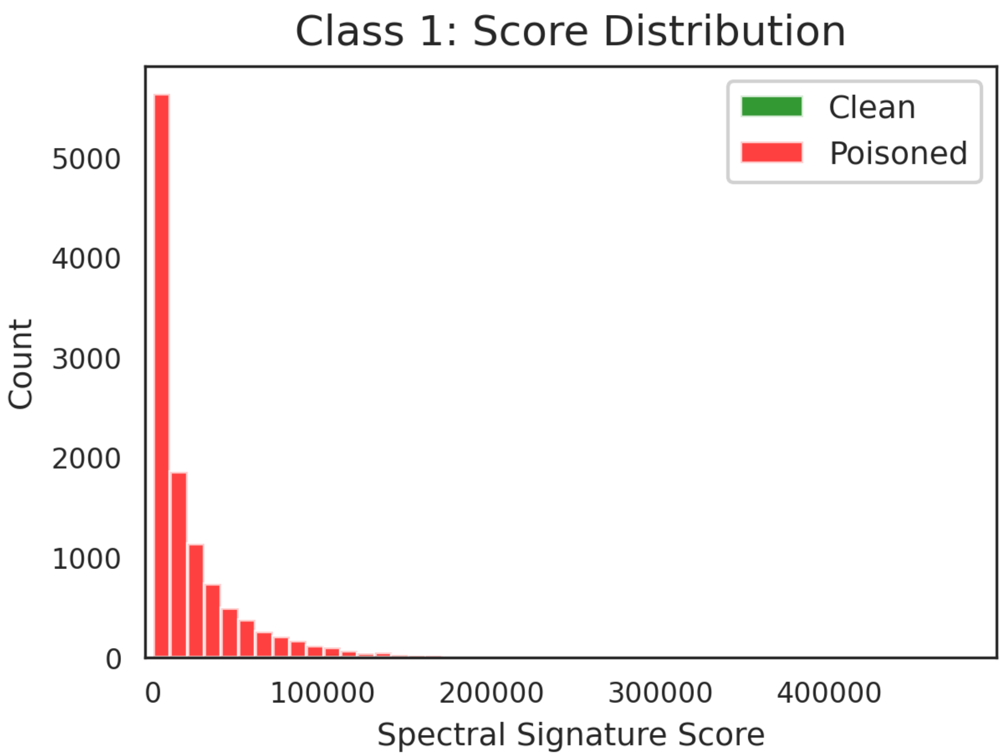

Figure 14: Spectral Signatures: target class=1, Model: Saint, Dataset: Credit Card, $\epsilon = 0.05$, and $\mu = 1.0$).

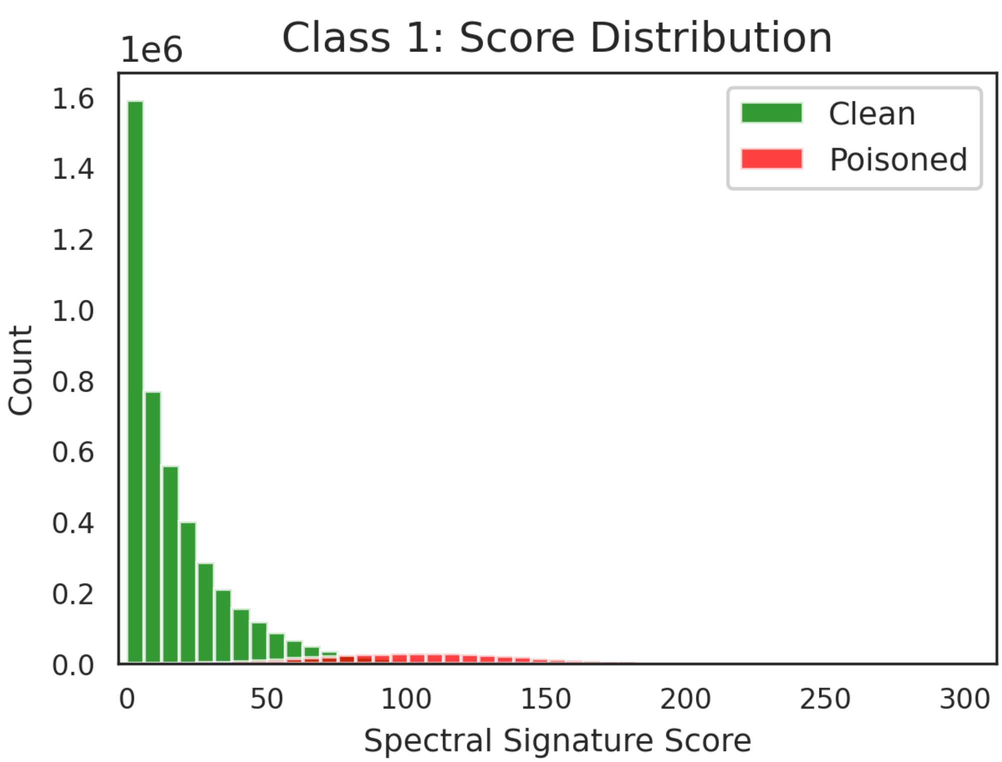

Figure 15: Spectral Signatures: target class=1, Model: Saint, Dataset: HIGGS, $\epsilon = 0.05$, and $\mu = 1.0$).